detect a limited number of gases due to which must be facilitated by adding compounds to different gases for humans to be able to detect it [2].

These limitations of human olfactory system make it difficult to rely on humans for the job of odor detection in Industries. Moreover, the odor detection of dangerous gases, even though possible by humans, may be fatal. An alternate approach is to train and utilize dog for odor sensing. This too has limitations as it is expensive to train dogs, and their life span is short and limited. These limitations have led to the development of electronic noses which try to mimic the human olfactory system. Electronic noses have proved their significance in various fields of health and industries and have been used as sensors for detection of food spoilage and in diagnosis of various diseases and much more [2]. Despite advances in the hardware of electronic noses, there hasn't been much attention paid to the software side of electronic noses.

This research aims to develop a general purpose Artificial Neural Network that can be used in various kinds of application from differentiating between markers, detecting food spoilage and diagnosing diseases.

## II. MODELS OF THE OLFACTORY SYSTEM

The objective of a great part of the examination with respect to the olfactory framework is to see how singular smells are recognized. Numerous analysts have created numerical models of the olfactory framework. These models regularly incorporate recreations of the neurobiological data preparing frameworks (biological neural networks). The olfactory data is handled in both the olfactory bulb and in the olfactory cortex. Figure 1 demonstrates the fundamental data preparing structures inside the brain. The olfactory cortex performs design grouping and acknowledgment of the detected smells. Once recognized, scent data is transmitted to the hippocampus, limbic framework and the cerebral cortex. The connection to the hippocampus explains why odor can sub-consciously evoke memories. Conscious perception of the odor and how to act on the odor takes place in the cerebral cortex [2]. The mammalian olfactory system uses a variety of chemical sensors, known as olfactory receptors, combined with signal processing in the olfactory bulb and automated pattern recognition in the olfactory cortex of the brain.

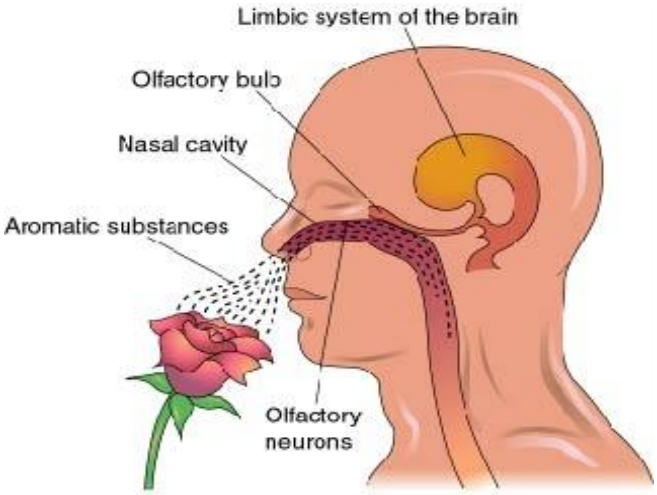


***Figure 1: The major processes of the olfactory system***

## III. ELECTRONIC NOSES

Electronic noses [3], [4] and [5], are being created as frameworks for the automated recognition and characterization of smells, vapours and gasses. The two principle parts of an electronic nose are the odor detection framework and the computerized pattern recognition framework. The odor detection framework can be an array of several different sensing elements (e.g., chemical sensors), where each element measures a different property of the sensed odor, or it can be a single sensing device (e.g., spectrometer) that produces an array of measurements for each odor, or it can be a combination of both chemical sensors and a spectrometer. By showing a wide range of smells to the sensor array, a database of signatures is developed. This database of marked odor signatures is utilized to train the pattern recognition system. The objective of this preparation procedure is to arrange and configure the recognition system to deliver extraordinary mappings of every odor so that an automated distinguishing proof can be executed [6].Albeit every sensor is intended for a particular chemical, each reacts to a wide assortment of chemical vapours. , On the whole these sensors react with one of a kind signature to various chemicals. Amid the preparation procedure, different chemicals with known blends are exhibited to the system.

## IV. ARTIFICIAL NEURAL NETWORKS AND ELECTRONIC NOSES

ANNs are widely used to utilize and examine different data representation methods such as complex data and pattern recognition. These techniques are indicating promising outcomes in chemical vapor recognition. Figure 2 illustrates an electronic nose prototype with the help ANN to identify odors from several common chemicals.


ISBN: 978-81-946255-9-9

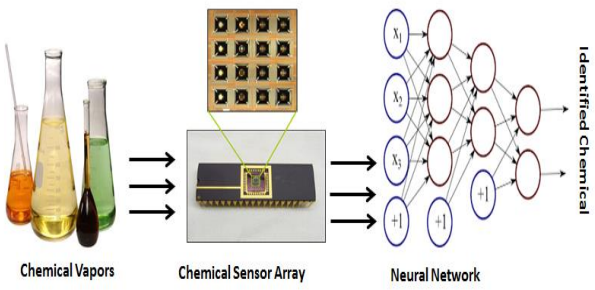


***Figure 2: Electronic nose Prototype Using ANN***

## V. RELATED WORK

Reviewing related works in the field of electronic nose and its use in odor sensing shows that various smell sensors have been used for the different purposes alongside different methods of analysis which were mainly statistical in nature and not artificially intelligent. Moreover, the trend of using custom smell sensors is also seen, with many being developed for very specific purposes to detect specific odors only. An overview of this is given in the following paragraphs.

***Research Paper***: Lung cancer identification by the analysis of breath by means of an array of non-selective gas sensors [7]. ***Smell Sensor Used:***LibraNose. ***Method/Algorithm for Odor Sensing:*** Partial Least Squares Discriminant Analysis. ***Accuracy:***94%.

***Research Paper***:Use of an electronic nose to diagnose bacterial sinusitis [8]. ***Smell Sensor Used:***Cyranose 320. ***Method/Algorithm for Odor Sensing:***Support Vector Machine. ***Accuracy:*** 74%

***Research Paper***:Predicting Type 2 diabetes using an electronic nose-based artificial neural network analysis [9]. ***Smell Sensor Used:***Custom Smell Sensor. ***Method/Algorithm for Odor Sensing:***Artificial Neural Network. ***Accuracy:*** 92%.

***Research Paper***:Detecting quality of indoor-air using electronic nose [10] . ***Smell Sensor Used:***Custom Smell Sensor. ***Method/Algorithm for Odor Sensing:***Fuzzy Logic Pattern Recognition. ***Accuracy:***90%-100%.

***Research Paper***:Application of ANN with extracted parameters from an electronic nose in cigarette brand identification [11]. ***Smell Sensor Used:***Cyranose 320. ***Method/Algorithm for Odor Sensing:***Artificial Neural Network. ***Accuracy:***80-100%

***Research Paper***:An electronic nose system to diagnose illness [12]. ***Smell Sensor Used:***Fox 2000. ***Method/Algorithm for Odor Sensing:***Statistical Analysis. ***Accuracy:***80.6%.

***Research Paper***:An investigation on electronic nose diagnosis of lung cancer. Lung Cancer [13]. ***Smell Sensor Used:***Custom Smell Sensor. ***Method/Algorithm for Odor Sensing:***Partial Least Squares Discriminant Analysis. ***Accuracy:***85.7%.

***Research Paper***:Electronic nose prediction of a clinical pneumonia score: biosensors and microbes[14]. ***Smell Sensor Used:***Custom Smell Sensor. ***Method/Algorithm for Odor Sensing:***Statistical Analysis. ***Accuracy:*** less than 50%.

Reviewing the aforementioned researches, this research uses a general-purpose sensor in form of OMX-GR and Artificial Neural Network for analysis due to its ability of being trained and being general purpose.

## VI. DATA SAMPLING

A handheld OMX-GR odometer sensor (Figure 3) is used in this research study to collect data samples for three different materials: White Board Marker, Muscle Cream and Herbal Inhalant (these materials are used because of their strong smell). The sensor includes two semiconductor gas senor with thecapability to detect a variety of odors from combustible gases. This sensor measures odor with different strength and classifications. The odor strength range from 0 to 999 and a classification feature range from 0 to 89. Additionally the sensor supports two types of sampling measurement modes: real-time sampling and memory samplings.

***Figure 3: OMX-GR Odometer Sensor and the Materials used for Data Sampling***

During this phase of the research work we measured approximately 1800 data representing the odor classification and strength of three different materials. For each material 15 data samples are collected in an interval of 20 seconds each. Table 1 describes a sample of data collected for each material representing the odor classification and strength of the material.


ISBN: 978-81-946255-9-9

| White Board Marker | Muscle Cream | Herbal Inhalant |
|---|---|---|
| 88,564 | 65,143 | 69,528 |
| 88,574 | 65,142 | 69,528 |
| 88,585 | 65,142 | 69,528 |
| 89,585 | 65,142 | 69,528 |
| 89,595 | 65,142 | 69,528 |
| 89,595 | 65,142 | 69,528 |
| 89,595 | 65,142 | 69,527 |
| 89,595 | 65,142 | 69,527 |
| 89,585 | 65,142 | 69,527 |
| 89,585 | 65,142 | 69,527 |

***Table 1: Sample data representing the classification and strength of odors measured***

Table 2 describes the range of the classification and strength of the entire data collected.

| White Board Marker | | | Muscle Cream | | Herbal Inhalant | |
|---|---|---|---|---|---|---|
| Sample # | Class | Strength | Class | Strength | Class | Strength |
| 1 | 89-89 | 889-906 | 65-66 | 138-152 | 71-72 | 481-502 |
| 2 | 89-89 | 857-906 | 64-65 | 138-152 | 71-71 | 502-522 |
| 3 | 89-89 | 826-873 | 64-66 | 143-157 | 70-71 | 522-524 |
| 4 | 88-88 | 640-769 | 65-66 | 147-157 | 69-70 | 525-526 |
| 5 | 88-88 | 650-694 | 65-66 | 138-147 | 69-69 | 526-527 |
| 6 | 87-88 | 592-650 | 65-65 | 138-142 | 69-69 | 526-527 |
| 7 | 88-89 | 564-595 | 64-65 | 148-162 | 68-69 | 510-523 |
| 8 | 88-89 | 535-574 | 65-65 | 162-163 | 68-68 | 510-511 |
| 9 | 88-88 | 508-544 | 65-65 | 157-162 | 68-68 | 510-511 |
| 10 | 88-88 | 446-476 | 65-66 | 162-167 | 68-68 | 511-511 |
| 11 | 87-88 | 412-446 | 65-66 | 162-167 | 68-68 | 511-511 |
| 12 | 86-87 | 387-412 | 64-65 | 152-162 | 68-68 | 511-511 |
| 13 | 88-89 | 703-729 | 64-65 | 148-162 | 68-68 | 511-511 |
| 14 | 88-89 | 611-691 | 65-66 | 162-167 | 67-68 | 493-512 |
| 15 | 88-88 | 553-611 | 65-66 | 158-167 | 67-68 | 493-494 |
| All Data | 86-89 | 387-906 | 64-66 | 138-167 | 67-72 | 481-528 |

***Table 2: The range of the measured odor classification and strength***

In the second phase of the data collection process, all of the 15 samples of the data for the three materials were graphed to identify the similarities and the overlapping of the samples before feeding it to the ANN. These similarities and overlapping is illustrated in Figure 4.

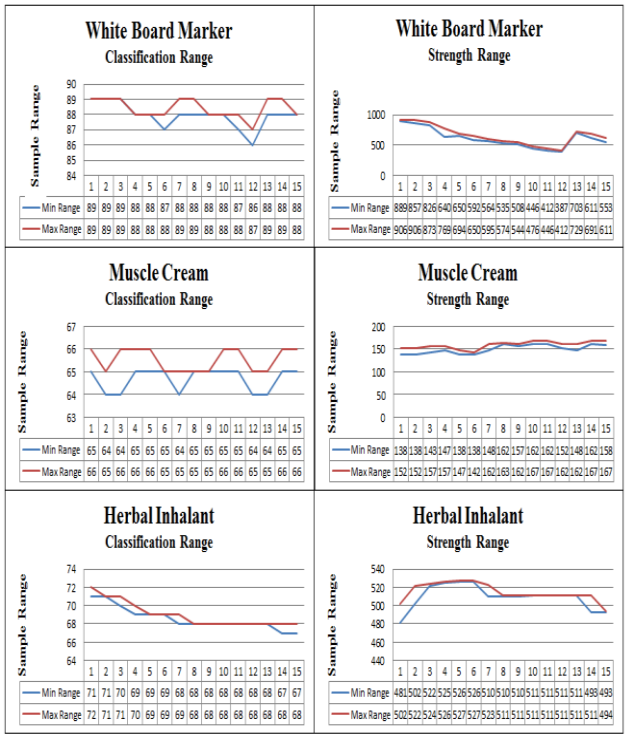


***Figure 4: Similarities and overlapping of the odor data samples for each material***

## VII. METHODOLOGY AND RESULTS

The two main components of an electronic nose are the sensing system and the automated pattern recognition system. In this research study, the sensing system is a pair of semiconductor gas sensor to classify odors. This is a popular simplified handheld OMX-GR odor meter used as a tool for odor analysis to enable the indication of relative strength and odor classification numerically by comparing odor gases and purified air. The sensor produced an array of measurements for three different materials (White Board Marker, Muscle Cream and Herbal Inhalant). Each material vapor presented to the sensor array produced a signature or pattern characteristic of the vapor as shown in table 1. To generalize the ANN, the data is divided into three sets: training set 50%, validation set 25% and testing set 25%.

A backpropagation supervised learning technique is used and resulted with 0.0999 training error, 0.05312 validations and 0% testing. A total of 3178 epoch was used. Figure 6 shows the setup to use sigmoid function as a nonlinear with S shape curved function. The figure then illustrates the training graph and main results generated after feeding the data to the ANN program.


ISBN: 978-81-946255-9-9

**b. Fitted Regretion Line**

Equation (1) presents the true regression line which is usually never known. However, the regression line can be estimated where estimating the coefficients $\beta_1$ and $\beta_0$for an observed data set. [6-7]

$$E(Y) = \beta_0 + \beta_1 x \quad (1)$$

Equation (2) represents the actual values of y which are assumed to be the sum of the mean value, E(Y) and a random error term ε.

$$Y = E(Y) + \epsilon \quad (2)$$
$$= \beta_0 + \beta_1 x + \epsilon$$

The least square estimates, $\hat{\beta}_0 \hat{\beta}_0$ and $\hat{\beta}_1 \hat{\beta}_1$ are obtained

from using the following equations:

Equations (3), (4) represent the least square estimates $\hat{\beta}_1$and $\hat{\beta}_0$ respectively:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \quad (3)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (4)$$

Where $\bar{x}$ is the mean of all predictor variable calculated

using equation (5), and $\bar{y}$ is the mean of all observed values

calculated using equation (6):

$$\bar{x} = (1/n) \sum_{i=1}^{n} x_i. \quad (5)$$

After know $\hat{\beta}_1 \hat{\beta}_1$ and $\hat{\beta}\hat{\beta}_0$ , the fitted regression line will be

written as:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \quad (6)$$

Where the difference between the corresponding observed value, $y_i$ and the fitted value, $\bar{y}_i$ is called the residual $e_i$:

$$e_i = y_i - \hat{y}_i \quad (7)$$

**c. Calculation of the statistique $F_0$**

The statistic $F_0$ test the significance of regression is calculated as follows:[6-7]

$$F_0 = \frac{MS_R}{MS_E} \quad (8)$$

Where: MSR is the regression mean square and MSE is the mean square error.

To calculate the statistic $F_o$, it must study the following six models [3][4][5]

**c.1 Total Sum of Squares ($SS_T$) model**

The $SS_T$ is obtained using the equation (9):

$$SS_T = \sum_{i=1}^{n} (y_i - \bar{y})^2 \quad (9)$$
$$= \sum_{i=1}^{n} y_i^2 - \frac{(\sum_{i=1}^{n} y_i)^2}{n}$$

**c.2 Sum of Squares regression ($SS_R$) model**

The $SS_R$, can be obtained using the equation(10):[5]

$$SS_R = \sum_{i=1}^{n} \hat{y}_i^2 - \frac{(\sum_{i=1}^{n} y_i)^2}{n} \quad (10)$$
$$= \hat{\mathbf{y}}'\hat{\mathbf{y}} - (\tfrac{1}{n})\mathbf{y}'\mathbf{J}\mathbf{y}$$
$$= \mathbf{y}'\left[\mathbf{H} - (\tfrac{1}{n})\mathbf{J}\right]\mathbf{y}$$

**c.3 Sum Squares Error (SSE)operation**

The $SS_E$ is obtained using the equation (11):

$SSE=SS_T-SS_R$ (11)

**c.4 The total Mean Squares ($MS_T$) model**

The ($MS_T$) are obtained by dividing the $SS_T$ with their associated degrees of freedom. The number of degrees of freedom associated with $SS_T$ is n-1 since there are an 'n' observations, but one degree of freedom is lost in the calculation of the sample mean ȳ.

$$MS_T = \frac{SS_T}{n-1} \quad (12)$$

**c.5 The regression mean square ($MS_R$) model**

The number of freedom degrees associated with the $SS_R$, is k. so, There are k+1 degrees of freedom associated with a regression model with k+1 coefficients $\beta_0,\beta_1,..\beta_k$. However, one degree of freedom is lost because the deviations ($\hat{y}_i$-$\bar{y}$)$(\hat{y}_i - \bar{y})$, are subjected to the constraints that they must

sum to zero $\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$

The $MS_R$ is obtained using the equation (13):

$$MS_R = \frac{SS_R}{k} \quad (13)$$

Since there are n observations in all, The number of degrees of freedom associated with the error sum squares is: n-(k+1), , but (k+1) degrees of freedom are lost in obtaining


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